Karnan Thamilchelvan

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SKILLS

Programming
C++, C, Java, Python, SystemVerilog, VHDL, MATLAB, SQL

Software
EasyEDA, KiCAD, SolidWorks, STM32Cube, Quartus, Flutter, Jira, Arduino IDE, VSCode

Hardware
PCB Design, Soldering/ PCB Rework, Circuit Debugging

Equipment
Oscilloscope, Function Generator, Multimeters

EXPERIENCE

Robotics & Controls Designer, *Jet Automation* — Jan 2026 – Apr 2026 | Mississauga, ON

- Developed an automated packaging system using an Omron TM12 collaborative robot; programmed robotic motion paths in TMFlow for box pickup, opening, folding, and taping across 4 box sizes with 10–15 second cycle times.
- Integrated Omron Sysmac PLC controls with 4 industrial sensors, vacuum suction end-effectors, and robot I/O, enabling reliable sensor-driven automation and coordinated production sequencing.
- Designed and deployed a custom HMI in Sysmac Studio for box-size selection, automation controls, industrial I/O monitoring, and emergency-stop integration through PLC mapping.
- Conducted Factory Acceptance Testing (FAT), troubleshooting, and firmware/hardware validation on robotic automation systems and forklift charging nodes.

Hardware & Embedded Systems Intern, *University of Waterloo - IDEAs Clinic* — May 2025 – Aug 2025 | Waterloo, ON

- Designed a Mechatronic Resonance System with Arduino UNO R4 WiFi, motion, proximity sensors, and a custom PCB; created schematics in EasyEDA, fabricated 50+ boards, and soldered components for reliable system integration in ECE198 (Dynamic Systems Lab).
- Modeled in SolidWorks and fabricated 220+ 3D-printed and machined parts, assembling 40 complete units that are now used to teach resonance and applied circuitry.
- Created and delivered FPGA programs on Quartus, providing beginner-to-intermediate students with hands-on exposure to digital design and embedded systems concepts.

Hardware Engineer, *Fallyx* — Sep 2024 – Dec 2024 | Toronto, ON

- Designed and developed custom PCBs for an IoT-enabled fall detection device deployed in 50+ retirement homes, integrating ESP32 microcontrollers and barometric pressure sensors for real-time monitoring and wireless caregiver alerts.
- Optimized fall detection hardware by integrating and calibrating Inertial Measurement Unit (IMU) sensors and refining signal processing, improving detection accuracy and system reliability.
- Prototyped a 3D-printed enclosure in SolidWorks, creating an ergonomic, durable design that enhanced usability as a medical-assistive device.

PROJECTS

RainSense Smart Water Management, *ESP32, Arduino IDE, Flutter*

- Built an ESP32-based weather station to solve a real-world problem on my parents' farm, reducing water bills by optimizing rainwater collection using real-time weather API data and automated irrigation schedules.
- Developed a Flutter mobile app to provide my parents with rainstorm alerts and water tank monitoring, integrating a distance sensor for capacity tracking and enabling smarter water usage and cost savings. Presented at Hack the North.

Smart-Irrigation Controller, *STM32, STM32CubeIDE, C, KiCad*

- Developed an automated irrigation system for terraced vineyards, integrating PWM-controlled pumps, servo-driven water distribution, and ultrasonic sensors to optimize water use based on real-time reservoir levels.
- Designed a custom PCB with an STM32 microcontroller and timer board for precise motor control and monitoring.

EDUCATION

University of Waterloo, — Sep 2022 - Apr 2027 | Waterloo, ON

Candidate for Bachelor of Applied Science – BASc, Computer Engineering

Relevant Courses: Digital Hardware Systems, Algorithms and Data Structures, Compilers